

BAHARI OPS

Mwani mlinzi AI

*An early warning and harvest
decision for Zanzibar's seaweed
farmers available on any phone.*



THE PROBLEM

Production is back. Value capture isn't.

Dried seaweed output grew from 19,716 t (2024) to 20,075 t (2025). Yet ~25,000 farmers about 80% women and still face climate, disease, quality and market-information risk with little support at the farm.

82.9%

keep no production records

~40%

dry seaweed directly on the ground

5.7%

show comprehensive knowledge of common diseases

94.3%

do no value addition

Ocean forecasts and expertise exist. What's missing is the step from information to a farm action.

PROBLEM STATEMENT

“How can we help Zanzibar's seaweed farmers to reduce avoidable crop losses and capture more value from each harvest despite warming seas, disease and no timely, farm-specific advice on the phones they already own?”

TARGET USERS

~25,000 seaweed farmers, about 80% women, organised in cooperatives across Pemba and Unguja

DESIRED OUTCOME

Fewer avoidable losses, better quality and complete farm records that open market access

MAIN BARRIER

Ocean risk and disease knowledge never reach the farm as a clear, timely action

TARGET USERS

Farmers use it; the value chain pays for it

THE USER

Seaweed farmers

~25,000 people, about 80% women, organised through cooperatives in Pemba and Unguja.

Most lack production records, disease knowledge and price information.

USSD + SMS on any basic phone - free or subsidised.

PAYER

Cooperatives

Farm records and harvest planning

PAYER

Extension services

Risk maps and prioritised field visits

PAYER

Programmes & NGOs

Measurable adoption and outcome data

PAYER

Processors & buyers

Forward supply visibility for procurement

MARKET OPPORTUNITY

A funded sector ready for a decision layer

25,000

seaweed farmers, about 80% women

20,075 t

dried seaweed output in 2025, up from 19,716 t

4.8%

of Zanzibar GDP from fisheries & aquaculture

US\$1.965M

UN/Joint SDG Fund Phase I for 15,000 households

WHY NOW

The UN programme (launched Aug 2025) is investing in cooperatives, value addition, finance and markets in N. Unguja and Pemba.

The Ministry is deploying SAMAKI digital monitoring and began a fisheries census in 2026.

Free Copernicus ocean data supplies the key environmental inputs.

WHERE IT GROWS

1. Seaweed risk & harvest prediction
2. Quality records
3. Co-op supply forecasting for buyers
4. Financing & insurance evidence
5. Other marine aquaculture, e.g. sea cucumber

THE SOLUTION

One farm-level question, answered every day

“Given my farm, crop stage and current ocean conditions, what should I do in the next 24–72 hours to reduce loss and improve the value of my harvest?”

Predict

Separate, explainable risk scores — heat & disease, storm & line damage, harvest window — from ocean signals, crop age and farmer reports.

Recommend

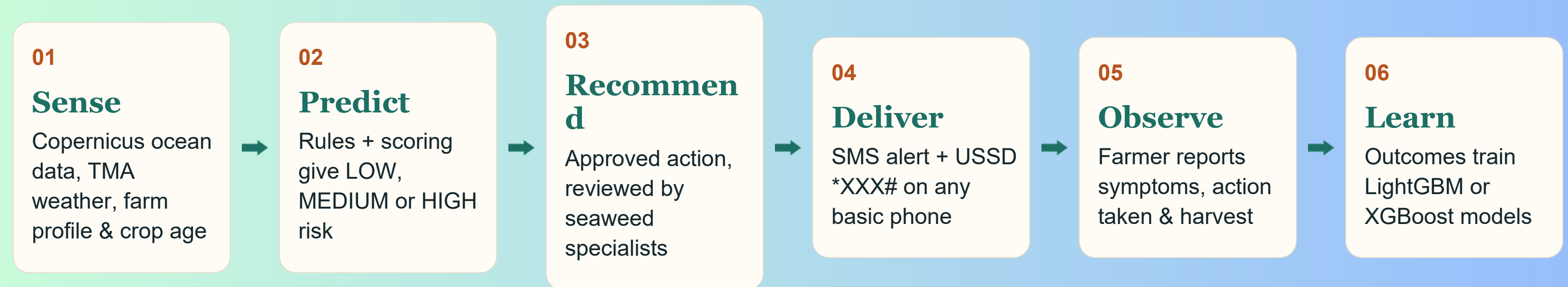
Actions come only from an extension-approved library — inspect, retie, harvest early. No LLM-invented farming advice.

Deliver

Short Kiswahili SMS and USSD for farmers; risk and supply views for cooperatives, extension officers and buyers.

HOW IT WORKS

A closed loop from ocean data to farm outcome



Every morning, 06:00–06:30: fetch ocean & weather data → build zone features → score risk → select approved actions → queue SMS. When a farmer dials USSD, the stored result returns instantly.

Every alert → action → outcome is recorded, so the model improves with each crop cycle.

KEY FEATURES

Six features that close the loop

Farmer & farm registry

Officers register farmer, GPS, variety and planting date on the dashboard.

Environmental pipeline

Copernicus & TMA signals per farm zone, precomputed every morning.

Risk classes → actions

Heat & disease, storm & line damage, harvest window — each mapped to an approved action.

Kiswahili SMS alerts

Proactive, persistent warnings. No smartphone or data plan needed.

USSD check & report

Status, alerts, symptoms, harvest kg & quality — confirmed before saving.

Co-op dashboard

Risk map, farmer reports, alerts, expected harvest and outcome records.

USSD · ANY GSM PHONE

Karibu MwaniMlinzi

1. Hali ya shamba
2. Tahadhari
3. Ripoti tatizo
4. Rekodi mavuno
5. Msaada

SMS: “Hatari ya joto ni kubwa katika eneo lako. Kagua mwani kesho asubuhi.”

BUSINESS MODEL

Free for farmers, paid by the value chain

Revenue stream	Customer	Pricing hypothesis
Programme / co-op licence	NGOs, donor programmes, co-ops	US\$8–12 per active farmer / yr
Buyer intelligence	Processors, exporters, aggregators	US\$300–600 / month
Institutional licence	Public programmes, large NGOs	US\$10k–25k / yr
Sponsored access	Buyers, banks, insurers, partners	Negotiated cohort fee
Risk data (later)	Insurers, lenders	Contract or revenue share

ILLUSTRATIVE CLUSTER

500 farmers × US\$10	US\$5,000
1 processor × US\$400 × 12	US\$4,800
Comms, data, hosting	– US\$1,750
Support & field servicing	– US\$2,000

≈62%

contribution margin (US\$6,050 on US\$9,800 revenue) before central costs

COMPETITIVE ADVANTAGE

Why MwaniMlinzi is hard to copy

Farm-specific

Risk tied to each farm's zone, crop age and observations — not one regional weather alert for everyone.

Explainable & safe

Calibrated risk plus a validated action library, with the reasons shown behind every alert.

Built for the last mile

USSD/SMS, no app or data bundle, cooperative-assisted onboarding.

No hardware to start

No IoT, drones or government operational databases needed for the MVP.

THE MOAT

A consented, longitudinal dataset linking **micro-location + ocean conditions + crop stage + recommended action + action taken + actual outcome**. Satellite data is open to everyone; this dataset is created only through use.

IMPACT

Pre-registered pilot targets for season one

$\geq 70\%$

farmers with usable production records

$\geq 75\%$

high-priority alerts acknowledged

$\geq 60\%$

alerts followed by the recommended action within 48 h

$\geq 70\%$

precision of “high risk” alerts once labels accumulate

$\geq 70\%$

farmers rating the advice useful or very useful

10–20%

fewer avoidable crop-loss incidents vs. matched control (experimental)

$< 25\%$

error on 7-day cooperative harvest forecasts

Down

ground drying: a substantial decline from participants' baseline

Win one cluster, then scale through partners

BEACHHEAD

1–2 cooperative clusters in Northern Unguja or Pemba

Where the UN/Joint SDG Fund programme already targets 15,000 seaweed households.

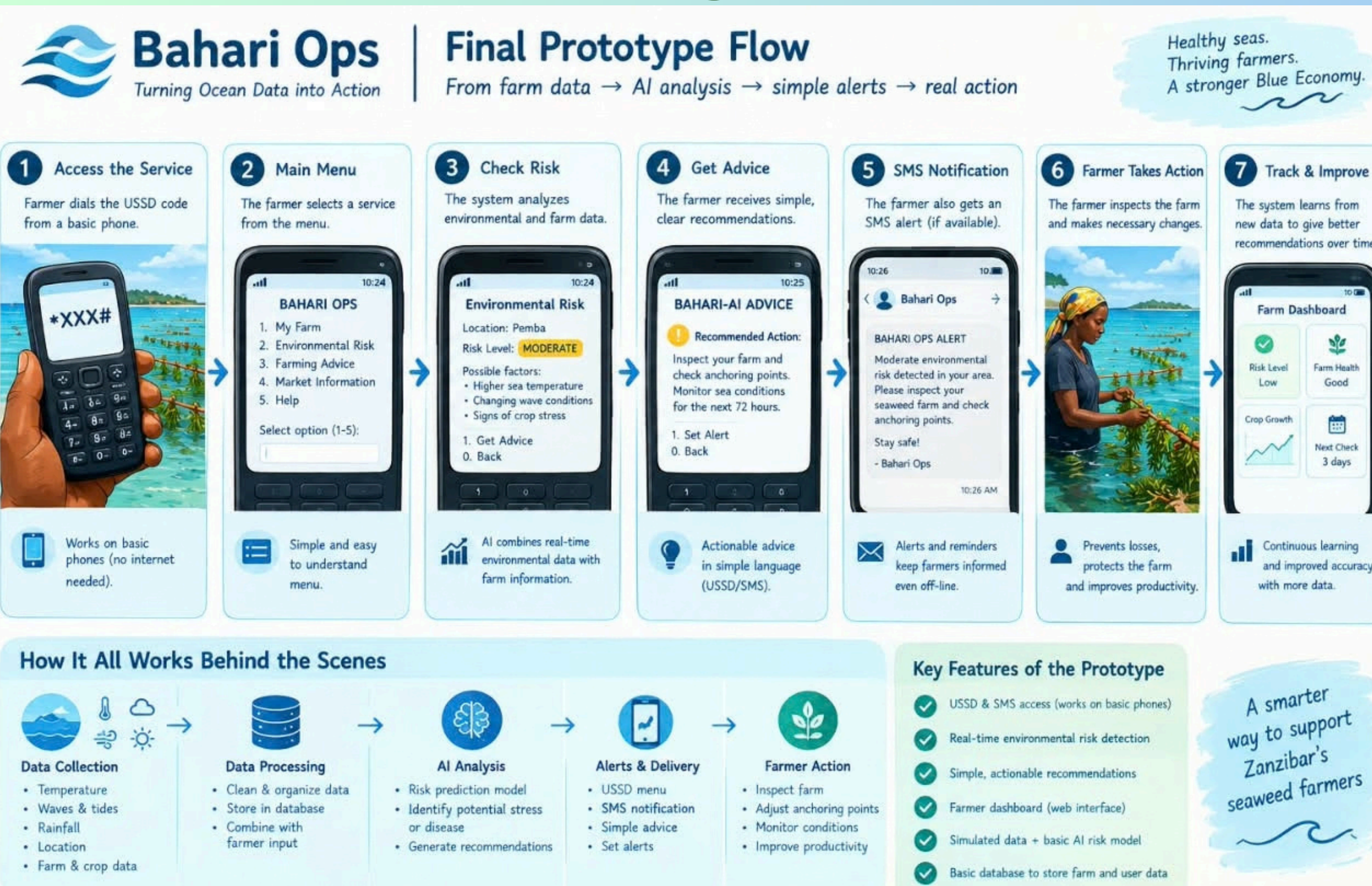
First pilot: 20–50 farmers · 1 cooperative · 1–2 risk types

Scale only once farmers repeatedly use and trust the service.

PARTNER STACK

Ministry of Blue Economy & Fisheries	Policy alignment, extension expertise, data interoperability
FAO/SWIOFC & UN Joint Programme	Pilot and validation partners
TMA	Authoritative weather information
Seaweed cooperatives	Trust, onboarding, ground truth, support
Processors & exporters	Demand signal; first private-sector payers
SUZA / Blue Economy Innovation Hub	Research, student talent, field testing
Africa's Talking / telcos	Distribution once the flow is validated

Our prototype



<https://claude.ai/artifact/WGLxksZ9hn3o3oee4QEo29>

Our Team

A multidisciplinary team focused on design and strategy

- SALHA SHAMIM KHAMIS - IT SPECIALIST
- SHEMSA NASSOR AMIN- SOFTWARE DEVELOPER
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